Audiophile - User Manual

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System Overview

Audiophile is a web-based application designed to help musicians improve their pitch accuracy and overall performance when playing. This system analyzes audio input, through live recording or file upload, and provides structured feedback based on pitch detection, accuracy, and consistency.

Technical Requirements and Specification

<u>Manual Deployment Steps</u>

1. **Create a Hugging Face Space**
    a. Go to huggingface.co and sign in.
    b. Click New Space, name it (e.g., AudioPhile), select Gradio as the SDK, and set visibility to Public.
2. **Upload all project files**
    a. Upload the following files on your Hugging Face Space:
        i. app.py
        ii. feedback.py
        iii. pitch_utils.py
        iv. instruments.py
        v. requirements.txt
        vi. styles.css
3. **Get a Groq API key**
    a. Go to https://console.groq.com/keys, sign in or create an account, and generate a new API key from the API Keys section.
4. **Add the API key as a Secret**
    a. In your Space, go to Settings → Secrets.
    b. Add a new secret:
        i. Key = GROQ_API_KEY
        ii. Value = your API key.
        iii. This keeps the key secure and out of your code.
5. **Restart the Space**
    a. Click Restart Space from the Settings tab.
    b. Hugging Face will install all dependencies from requirements.txt and launch the Gradio app automatically.

6. **Verify the deployment**
    a. Open the Space URL. The Gradio interface should load with instrument selection, age group, and audio input options visible.

Automatic Deployment Steps

1. **Duplicate the Space**
    a. Visit https://huggingface.co/spaces/MaxKoganUncc/Audiophile_Themes
    b. Click the three-dot menu (···) in the top right and select Duplicate this Space.
    c. Give it a name, set visibility, then follow Steps 3 - 6 from the Manual Deployment Steps to add your Groq API key and done.

Technical Specifications

System Requirements

Component	Requirement
Internet	Stable broadband connection to access Hugging Face Spaces
Microphone	Built-in or USB mic (required for live recording mode)
Audio files	In .wav or .mp3 format; formats supported by librosa

Technical Stack

Component	Technology
Frontend / UI	Gradio (Python web UI framework)
Hosting platform	Hugging Face Spaces
Pitch detection	librosa (for frame-by-frame analysis), numpy
Audio I/O	soundfile
AI feedback	Groq API (LLM-generated structured feedback)
Language	Python
Key files	app.py, feedback.py, pitch_utils.py, instruments.py, requirements.txt, styles.css

Main Features Explanation

Audiophile is designed to be accessible for a wide range of users while providing relevant and personalized feedback. The system's key goal is to help users better understand and improve their pitch accuracy when playing musical instruments using audio analysis and AI-generated guidance.

Instrument and User Customization

When starting the application, users are able to select both their instrument and age group. This information is used to provide personalized feedback that is accurate and helpful. Different instruments have different ranges of pitch, and the age-based adjustments are designed so that feedback is understandable to the user's skill level.

Flexible Audio Input

Audiophile supports two methods of audio input:

- Live recording using a microphone directly in the browser
- File upload

This flexibility allows users to either practice on the spot or analyze previously recorded audio.

Real-Time Pitch Detection and Analysis

Once the audio is submitted, the system processes and analyzes the input using pitch-detection algorithms. The system determines the following information:

- Detect the pitch of the performance
- Convert the audio into musical notation (such as A, C#, etc.)
- Determine if the pitch is in tune or flat

Accuracy Scoring

The system evaluates the score of the user's performance with the following two metrics:

- Pitch Accuracy: How close the notes are played to the correct pitch
- Pitch Consistency: How stable the pitch remains over the course of the performance.

To improve scoring accuracy, Audiophile analyzes the detected pitches using the information provided about the instrument selection. This prevents incorrect pitch detections and ensures that the score is relevant and accurate.

AI-Generated Feedback

After analysis, Audiophile uses an AI model, the Groq API, to generate easy-to-understand feedback. It will provide tips and suggestions to the user so that they can improve their performance. The analysis also rationalizes the "technical" side of the pitch detection algorithm into plain language.

Results Display

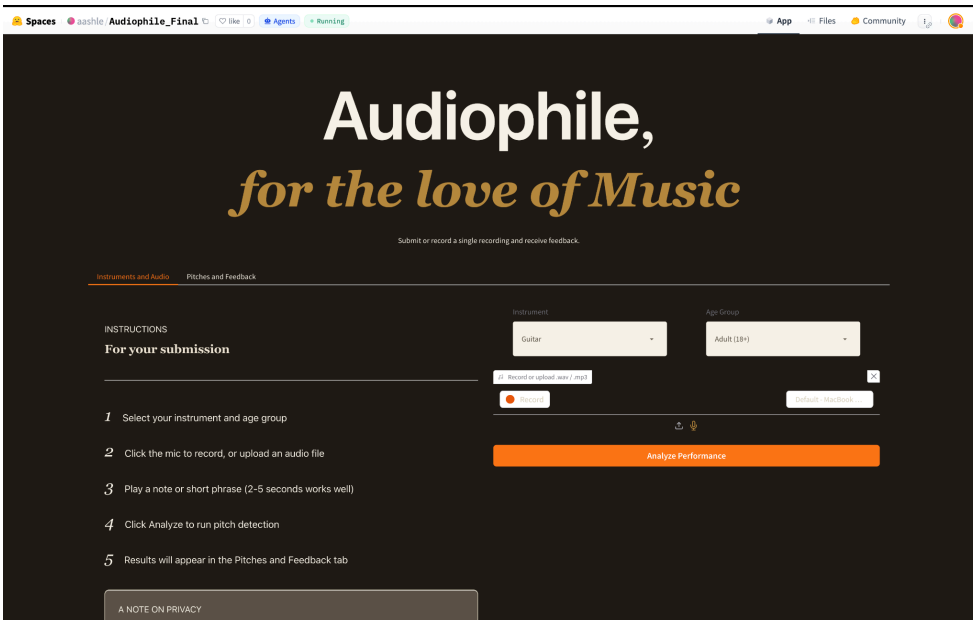
- Once the entire process is complete, all of the results can be seen on a single page that the user can access immediately. Users can view:
    - Detected notes and pitch deviations
    - Accuracy and consistency scoring
    - Personalized feedback and tips

Walkthroughs

Main Scenario

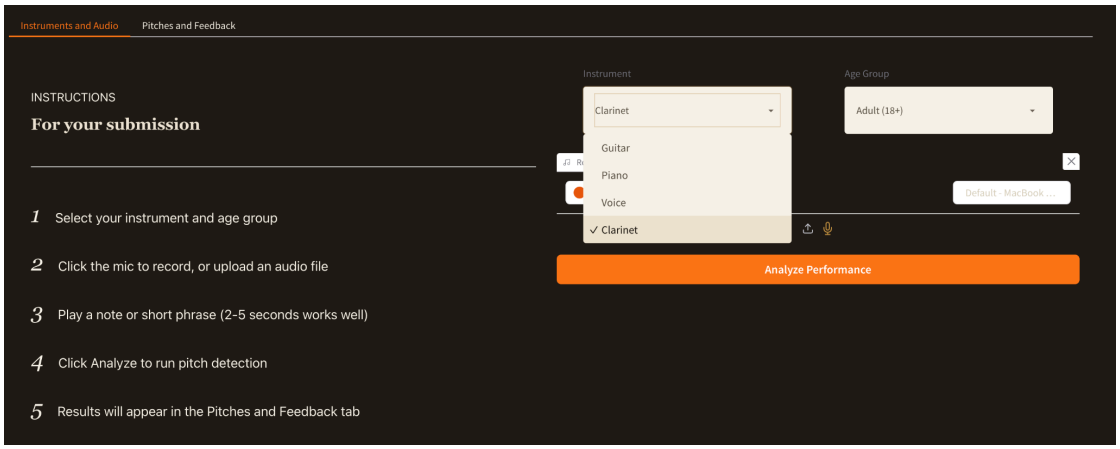
This scenario uses Audiophile for an adult Clarinet player by uploading an audio recording.

- **Open the Application**
  The user opens the Audiophile web application through the provided Hugging Face link.

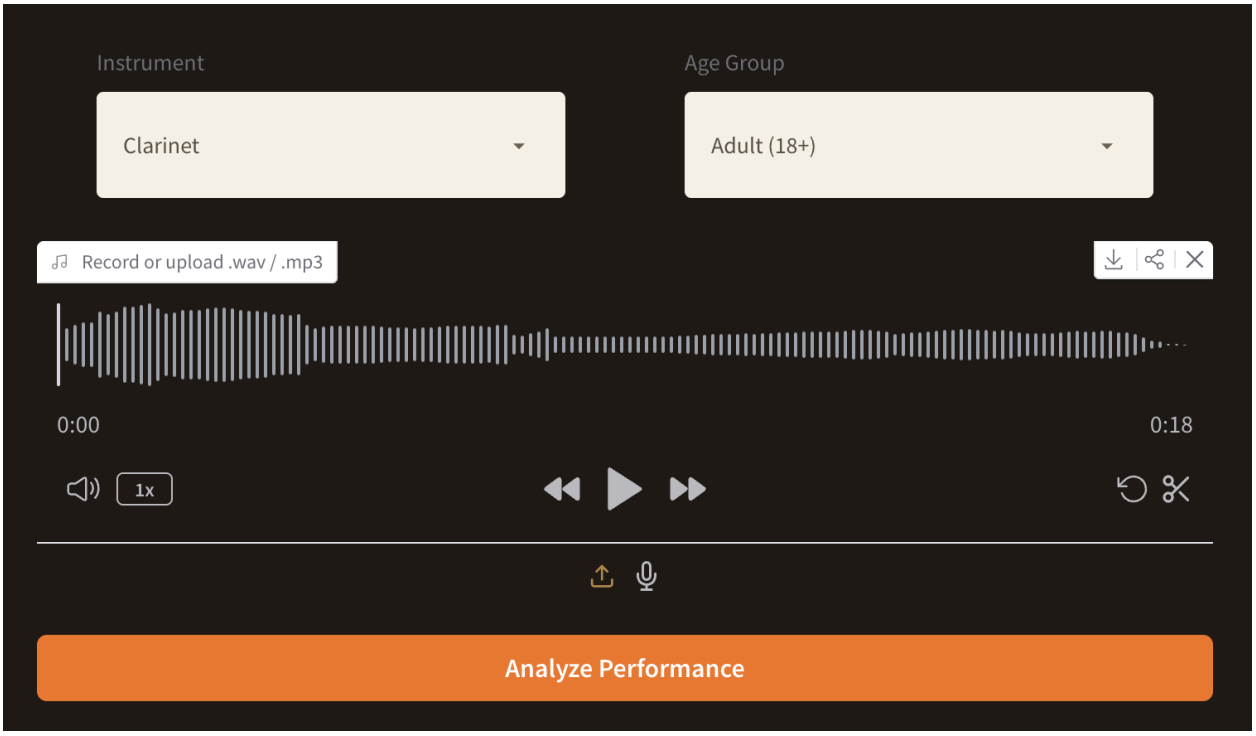


- **Select Instrument and Age Group**
  The user selects:
    - Their instrument (Guitar, Voice, etc.)
    - Their age group

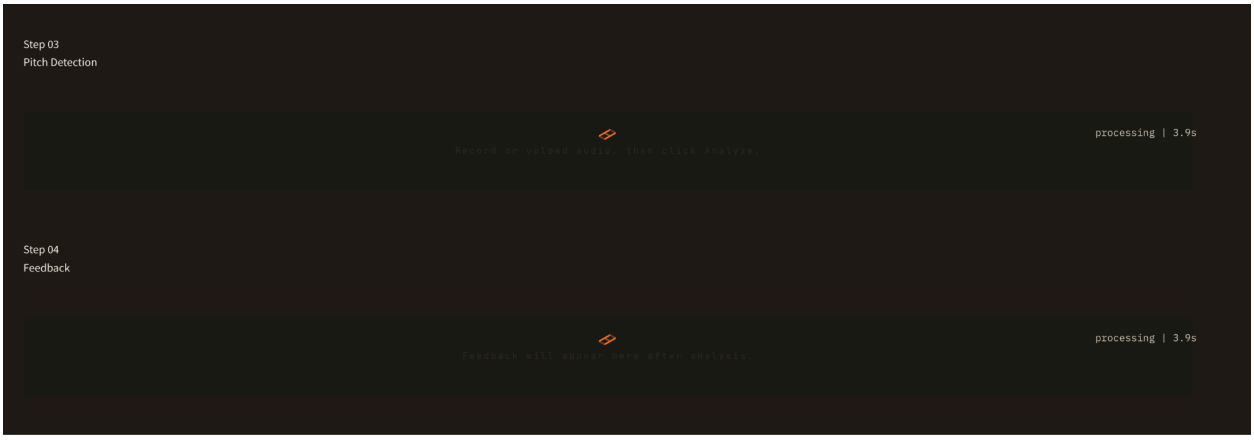
  This prompts the system so that pitch detection and feedback parameters are adjusted appropriately.



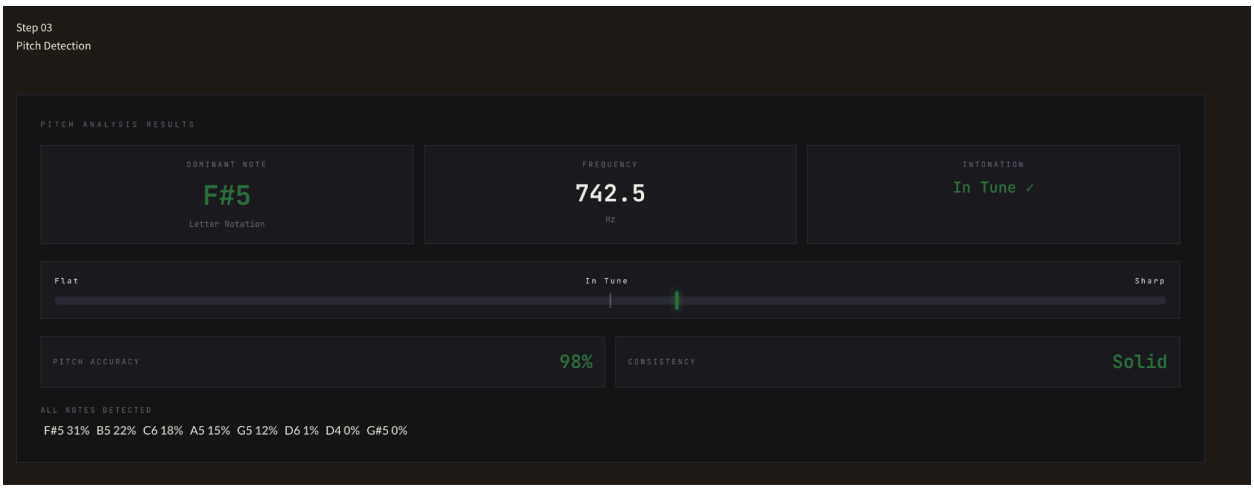
- **Upload Audio File**
  The user can either:
  - Record audio using the microphone
  - Upload an existing audio file

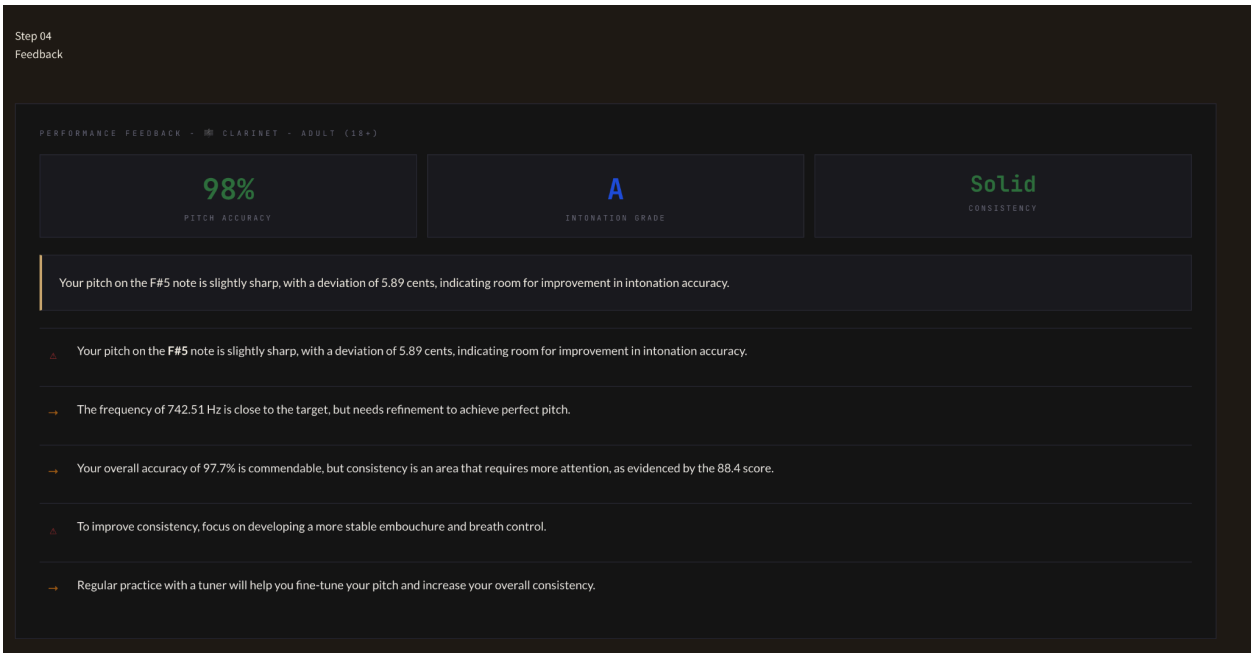


- **Analyze Performance**
  The user clicks the **“Analyze Performance”** button. The system then:
  - Processes the audio
  - Detects pitch
  - Calculates accuracy and consistency
  - **View Feedback**



- **View Results**
  After analysis, the system displays:
  - Detected note
  - Pitch deviation
  - Accuracy and consistency scores
  - AI-generated feedback

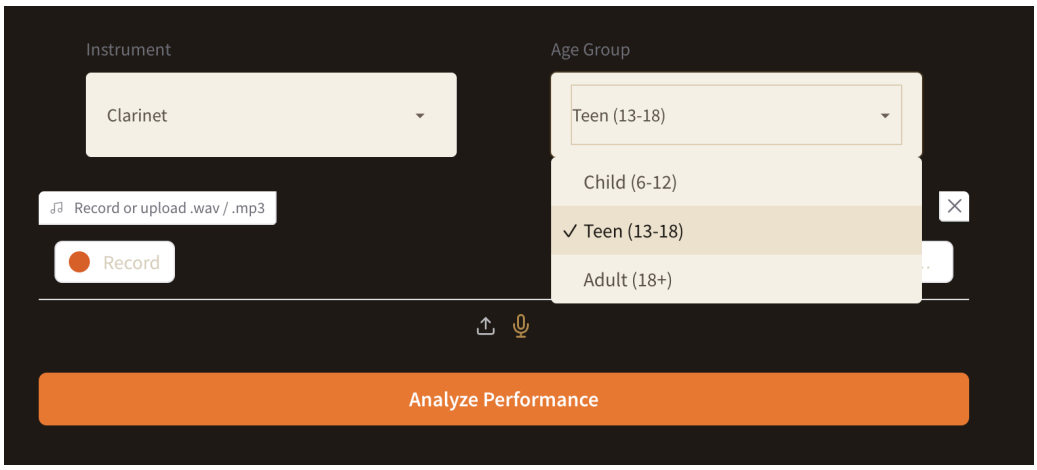




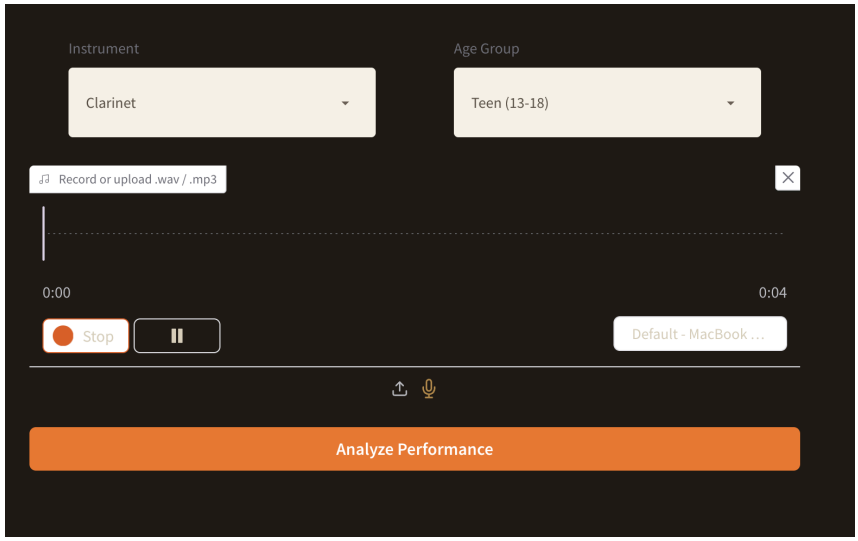
Additional Scenario 1

This scenario shows feedback for a Teen clarinet player and a live recording to analyze and get feedback from.

- **Open Hugging Face Space, and Select Options:**
  In this scenario, the following options are selected:
  - Instrument: Clarinet
  - Age Group: Teen (13-18)

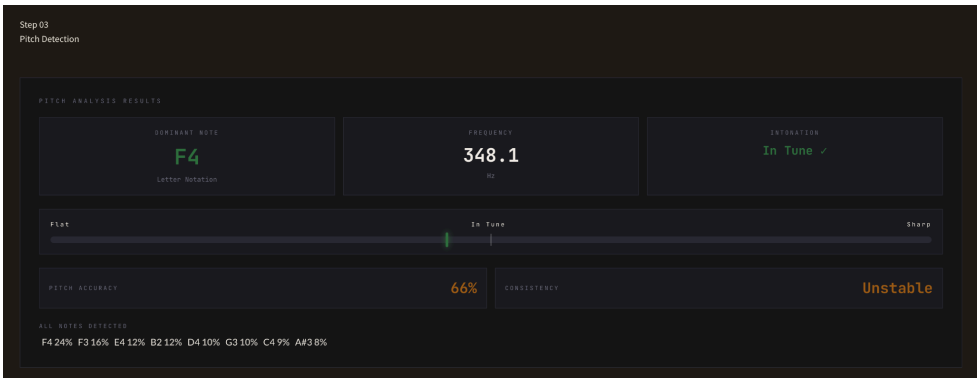


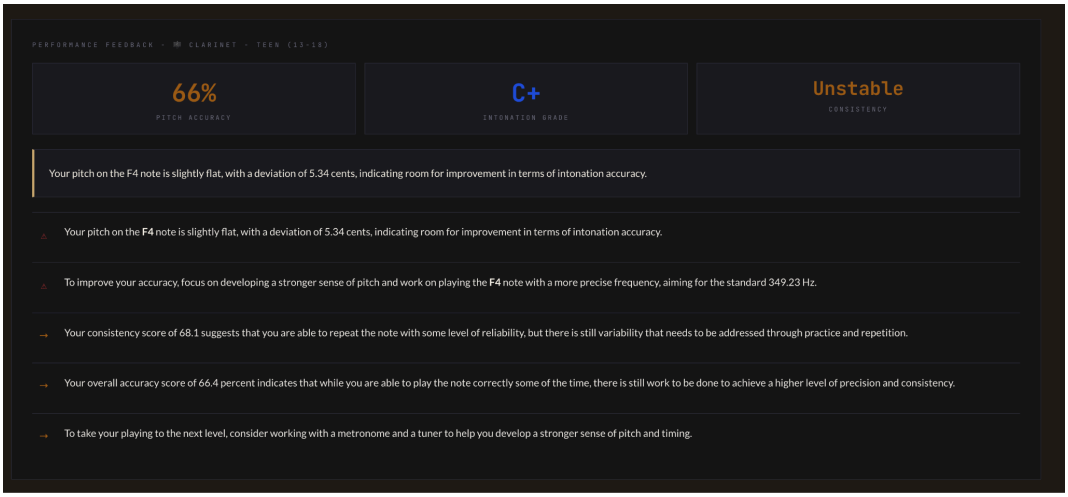
- **Press ‘Record’ and begin playing**
  An F Major scale is played for this live recording.



- **Press ‘Pitches and Feedback’**

If there was previous feedback, the page updates with new feedback relating to the new pitches that were detected in the live recording.

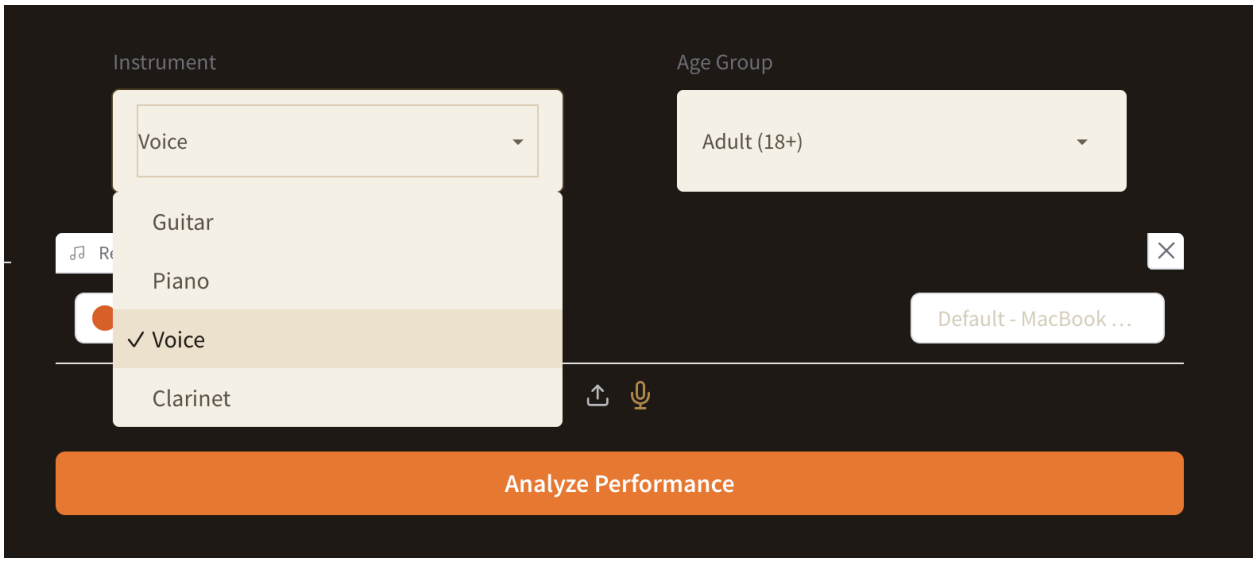




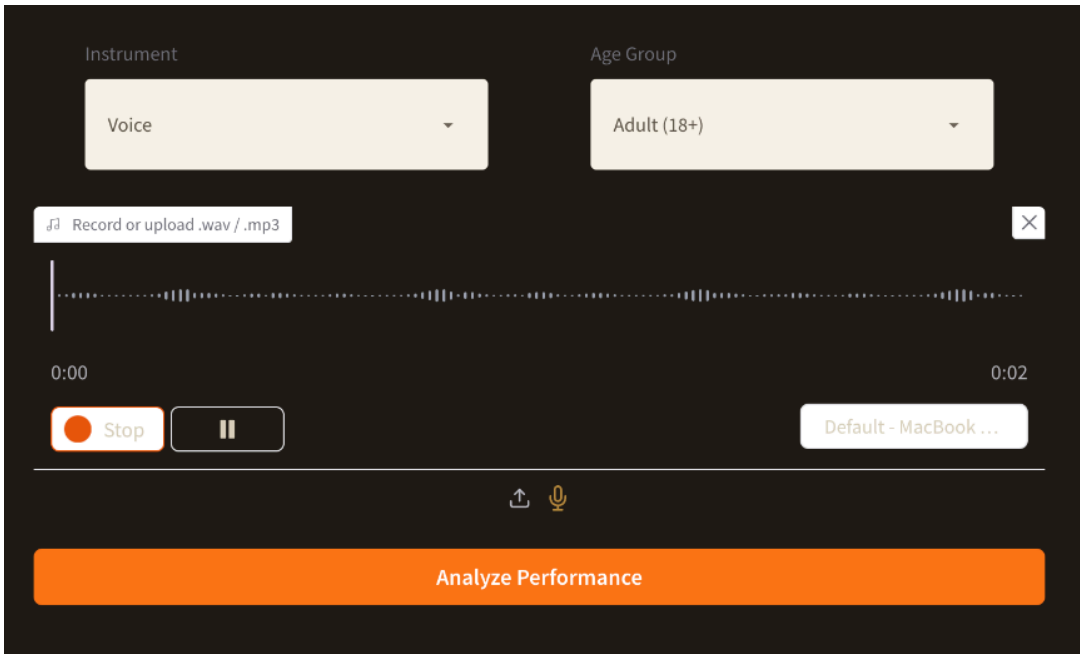
Additional Scenario 2

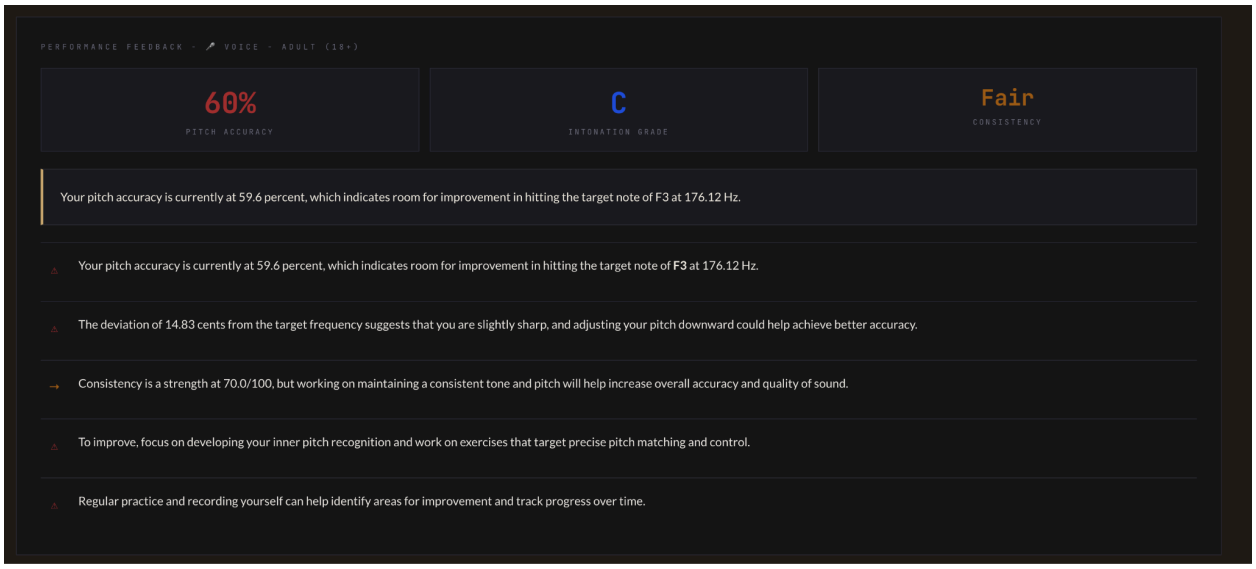
This second additional scenario for the Audiophile web application shows feedback for an adult Vocalist.

- **Open Hugging Face Space, and Select Options:**
  In this scenario, the following options are selected:
  - Instrument: Voice
  - Age Group: Adult (18+)



- **Press 'Record' and begin singing**
  Random pitches were sung by the tester for this live recording.



- **Press ‘Pitches and Feedback’**

Previous feedback is overwritten and new feedback is shown for the new live recording. Now, the user can take this feedback and potentially improve their sound by repeating the steps shown.